

FIN Local Research — Release 10.41

Checkpoint P454–P457

A Certified Nested Comb Dual, an Ordered-Symmetry Obstruction, and a Six-Decimal Global Cover

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Central result. An explicit rational nested-dual certificate and the inherited coherent primal witness confine the global three-slot causal half-distance to

$$0.52332810026048937 \leq D_3^{\text{global}} \leq 0.523334700252,$$

a rigorous gap below 6.6×10^{-6} .

Additional results. Known ordered-comb symmetries leave a five-dimensional fixed space and cannot force the three-dimensional coherent ansatz. A new globally licensed interval cover confines the coarse erasure optimum to a gap below 10^{-6} .

Boundary. No exact optimizer, optimizer uniqueness, selector, dimensional source, laboratory record, complete legacy-strict bridge, role transfer, or physical closure is claimed.

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Publication statement

This monograph reports local analytical and computer-assisted Programs P454, P455, and P457. It uses no laboratory record, external audit, internet result, remote computation, or physical validation.

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Chapter 1

Executive summary

This checkpoint completes local Programs P454, P455, and P457. It uses only analytical derivation, finite-dimensional linear algebra, rational arithmetic, directed intervals, and deterministic local optimization. It contains no laboratory data, external audit, internet result, remote computation, or physical evidence.

The three programs produce the following results.

1. **P454** — [**Proven / Computer-assisted proof**]. The complete three-slot causal discrimination problem from P451 is written as a primal semidefinite program and a nested dual with matrix sizes

$$8 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

A dependency-free log-determinant central-path scout locates a dual near the primal value. All entries are then rounded to explicit rationals. Six matrix slacks and two scalar slacks are certified strictly positive using exact Sylvester criteria. The two trigonometric slacks retain a margin greater than $4.9999999969 \times 10^{-7}$ after paying the full operator perturbation. Together with the inherited P451 coherent primal witness, this proves the global full-cone bracket

$$0.52332810026048937 \leq D_3^{\text{global}} \leq 0.523334700252.$$

Its exact width is

$$0.00000659999151063.$$

The corresponding success probability is confined to

$$0.761664050130244685 \leq p_{\text{succ}}^{\text{global}} \leq 0.761667350126.$$

2. **P455** — [**Proven / Strong evidence**]. The complex three-slot causal cone has 21 affine dimensions. Complex conjugation and global bit complement can be averaged because the objective is concave and invariant, so an optimizer has a real, complement-fixed representative. The real cone has 14 affine dimensions. Exact finite rank gives nine independent complement constraints and a five-dimensional fixed space. The O167 coherent ansatz has only three dimensions. Therefore two symmetry-allowed directions remain, and known symmetries alone cannot prove reduction to O167. This is a rigorous obstruction to the proposed symmetry shortcut. Eight deterministic searches over all five fixed dimensions return to the O167 value within 6.7×10^{-16} , with a negative sampled Hessian, but this non-improvement remains strong evidence rather than a theorem.
3. **P457** — [**Computer-assisted proof**]. P452 already licensed reversal reduction of the exact coarse erasure objective to the palindromic line. P457 performs a new outward-rounded cover at tolerance 10^{-6} . The cover evaluates 5,008 boxes and terminates with

$$0.463278283192093 \leq D_{\text{coarse}}^{\text{global}} \leq 0.46327928294340853,$$

whose width is

$$9.997513155113325 \times 10^{-7}.$$

This is a factor of approximately 992.7 narrower than the Release-10.40 bound.

The checkpoint constructs three typed objects:

- **O170 — Nested Comb Dual Ladder;**
- **O171 — Ordered-Comb Symmetry Residual Space;**
- **O172 — Six-Decimal Coarse-Erasure Cover.**

No exact P454 optimizer, optimizer uniqueness, selector, canonical physical orientation, dimensional unit, apparatus, laboratory record, complete legacy-to-strict bridge, role-transfer theorem, Standard Model, gravity theory, L_{total} , or Theory of Everything is claimed.

Chapter 2

Confidence convention

Label	Meaning
[Proven]	Exact analytic or finite algebraic result under declared premises
[Computer-assisted proof]	Finite rational/interval certificate with auditable code and artifacts
[Strong evidence]	Deterministic numerical agreement and falsification tests without a complete certificate
[Conditional]	Result depends on an explicitly supplied interface or variational rule
[Open]	Current artifacts do not decide the proposition
[Blocked by external evidence]	A laboratory, calibration, custody, or external record is indispensable

Chapter 3

Inherited assumptions and state map

3.1 Inputs retained from Release 10.40

- P449 gives the exact compressed causal recursion

$$\mathcal{C}_n = \{B_0 \oplus B_1 : B_0, B_1 \succeq 0, B_0 + B_1 \in \mathcal{C}_{n-1}\}.$$

- P451 gives the certified rational coherent primal witness with

$$D_{\text{primal}} \geq 0.52332810026048937.$$

- P452 proves reversal symmetrization of the declared actual coarse erasure objective, not merely its fine-label majorant.
- P446 provides the outward one-dimensional branch procedure reused in P457.
- P453 proves uniqueness of the minimum-negative-mass signed representation, conditional on its inherited certified inputs.

3.2 State map

Lane	State before this checkpoint	New state
Three-slot causal optimum	Primal lower 0.52332810026048937; full optimum open	Rigorous global upper 0.523334700252; gap $< 6.7 \times 10^{-6}$
Coherent symmetry face	Three-parameter O167 candidate	Five-dimensional exact fixed space; two residual directions proved
Coarse erasure code	Full-simplex gap $< 10^{-3}$	Full-simplex gap $< 10^{-6}$
P453 canonical gauge	Unique only inside minimum-TV rule	Unchanged
Selector/QW-2191	Open	Unchanged
Dimensional source	Missing	Unchanged
Laboratory evidence	None	None
Legacy-to-strict bridge	Incomplete	Unchanged

3.3 Kernel and ontology guardrails

The intermediate and strict kernels remain distinct:

$$K_{\text{legacy,ont}}(d) = \frac{\alpha_{\text{geo}} \cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d},$$

$$K_{\text{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}.$$

Kernel new Legacy / Legacy* remains a researched historical/intermediate bridge subclass and is not silently substituted for the strict kernel. The present programs are downstream reduced-channel calculations. They derive neither kernel and transfer no legacy physical role.

The internal ontology remains

$$\text{nadsoliton} \rightarrow \text{light} \rightarrow \text{matter} \rightarrow \text{emergent observer},$$

with no lower informational layer beneath the nadsoliton.

Chapter 4

Program P454 — exact nested comb dual and global bracket

4.1 Exact question

Can the full 21-dimensional three-slot causal optimization left open by P451 receive a matching rigorous upper bound without installing an external SDP package or treating a floating optimizer as a theorem?

4.2 Primal SDP

Let $\Delta = \Delta_3$ be the supplied eight-dimensional compressed process difference. Introduce positive measurement testers $T_+, T_- \succeq 0$ and positive causal-normalization layers

$$B_0, B_1 \in \mathbb{H}_+^4, \quad C_0, C_1 \in \mathbb{H}_+^2, \quad d_0, d_1 \geq 0.$$

The primal is

$$\begin{aligned} & \text{maximize} && \frac{1}{2} \text{Tr}[\Delta(T_+ - T_-)] \\ & \text{subject to} && T_+ + T_- = B_0 \oplus B_1, \\ & && B_0 + B_1 = C_0 \oplus C_1, \\ & && C_0 + C_1 = \text{diag}(d_0, d_1), \\ & && d_0 + d_1 = 1. \end{aligned}$$

For fixed normalization $N = T_+ + T_-$, binary Helstrom optimization gives

$$\max_{T_{\pm}} \frac{1}{2} \text{Tr}[\Delta(T_+ - T_-)] = \frac{1}{2} \|N^{1/2} \Delta N^{1/2}\|_1.$$

Thus the SDP is exactly the P451 full-cone objective, not a relaxation.

4.3 Dual derivation

Introduce Hermitian multipliers

$$X_3 \in \mathbb{H}^8, \quad X_2 \in \mathbb{H}^4, \quad X_1 \in \mathbb{H}^2,$$

and scalar λ . Taking the supremum over every positive primal variable is finite precisely when

$$X_3 \succeq \frac{\Delta}{2}, \quad X_3 \succeq -\frac{\Delta}{2},$$

$$X_2 \succeq (X_3)_{00}, \quad X_2 \succeq (X_3)_{11},$$

$$X_1 \succeq (X_2)_{00}, \quad X_1 \succeq (X_2)_{11},$$

$$\lambda \geq (X_1)_{00}, \quad \lambda \geq (X_1)_{11}.$$

Here the subscripts denote the two equal-size principal blocks at the next causal level. The dual is

$$\text{minimize } \lambda.$$

This produces the new hierarchy

$$\boxed{X_3 \rightarrow X_2 \rightarrow X_1 \rightarrow \lambda},$$

which is O170.

Both sides possess strict feasible points: the central normalized tester is strictly primal feasible after a strictly positive outcome split, and large scalar identities are strictly dual feasible. Finite-dimensional Slater duality therefore applies. Weak duality alone already suffices for the certified upper bound.

4.4 Dependency-free locator

No local cvxpy, cvxopt, SCS, Clarabel, or commercial SDP solver is available. P454 therefore uses the local barrier

$$\Phi_\mu = \lambda - \mu \sum_{j=1}^8 \log \det S_j,$$

where S_j are the six matrix and two scalar dual slacks. Complex conjugation swaps the first two constraints, so the dual variables can be chosen real symmetric while Δ remains imaginary Hermitian.

A deterministic feasible BFGS method with explicit Armijo backtracking follows twelve barrier values from 10^{-1} to 3×10^{-7} . The last point has

$$\lambda_{\text{scout}} = 0.5233347002511172.$$

The last barrier iteration does not meet the chosen gradient tolerance. It is used only to locate rational matrices and is never itself the certificate.

4.5 Rationalization and exact positivity

Every entry of X_3, X_2, X_1 is rounded to denominator 10^{12} , and λ is rounded upward:

$$\lambda_{\mathbb{Q}} = \frac{130833675063}{250000000000} = 0.523334700252.$$

The entries of $\Delta/2$ are enclosed by the inherited rational Machin/Taylor provider and rounded to denominator 10^{20} . The maximum entry radius is paid using

$$\|E\|_2 \leq 8 \max_{ij} |E_{ij}|.$$

For each rational midpoint matrix slack S , exact Sylvester tests are applied to

$$S - \frac{1}{2\,000\,000} I.$$

All leading principal minors are strictly positive:

- eight for each 8×8 trigonometric slack;
- four for each 4×4 inherited slack;
- two for each 2×2 inherited slack;
- one for each scalar slack.

The trigonometric operator perturbation is only

$$3.0514767686602317 \times 10^{-16},$$

so the certified exact minimum-eigenvalue lower bound for each top slack is

$$4.9999999969485233 \times 10^{-7} > 0.$$

The other matrix slacks retain the exact margin $1/2,000,000$, and the two scalar slacks are approximately 5.99997×10^{-7} and 5.99995×10^{-7} . Thus the rational dual is strictly feasible.

4.6 Global theorem and limitation

Combining the P451 primal lower bound and P454 dual upper bound gives

$$0.52332810026048937 \leq D_3^{\text{global}} \leq 0.523334700252.$$

The exact gap is

$$\frac{659999151063}{10^{17}} = 6.59999151063 \times 10^{-6}.$$

P454 verdict: the nested dual derivation is **[Proven]**, and the rational upper bound plus global bracket are **[Computer-assisted proof]**. Exact attainment by O167 and optimizer uniqueness remain **[Open]**.

Chapter 5

Program P455 — ordered-comb symmetry residual space

5.1 Exact question

Can the three-parameter O167 face be derived from the symmetries already present in the ordered three-slot cone, thereby converting its numerical optimum into the full optimum?

5.2 Real causal tangent space

The P449 complex Hermitian affine dimension is 21. Because

$$D(N) = D(\overline{N})$$

and D is concave, averaging N with $\overline{N}$ cannot lower the objective. Hence a real-symmetric optimizer exists.

The exact real affine recursion has dimension

$$e_1 = 1, \quad e_n = e_{n-1} + \frac{2^{n-1}(2^{n-1} + 1)}{2}.$$

For three slots,

$$e_3 = 1 + 3 + 10 = 14.$$

An explicit integer basis consists of one last-level diagonal direction, three real-symmetric C_0 directions, and ten real-symmetric B_0 directions.

5.3 Global complement

Let J_8 reverse all three history bits. The objective is invariant under the induced global-complement action. Averaging with

$$N \mapsto J_8 N J_8$$

again cannot lower a concave invariant objective.

On the exact fourteen-direction integer basis, the linear constraints

$$J_8 N J_8 = N$$

have rank nine. Therefore the real, complement-fixed affine space has

$$14 - 9 = 5$$

dimensions. The computation exports five exact nullspace vectors with entries in $\{-1, 0, 1\}$.

5.4 Why O167 is not symmetry-forced

The O167 form

$$N(A, B, u) = B_0(A, B, u) \oplus B_1(A, B, u),$$

with

$$C = \frac{1}{2} - A - 2B$$

has only three independent affine directions. Direct rank computation proves that this face is contained in the five-dimensional fixed space but does not fill it. The residual dimension is

$$5 - 3 = 2.$$

Thus complex conjugation plus global complement do not imply the O167 ansatz. Any proof of that reduction needs one additional invariant, a KKT theorem, a dual complementary-slackness argument, or an independent source law.

This is a rigorous obstruction, not a failed numerical search.

5.5 Falsification in all five dimensions

The O167 face optimization gives

$$(A, B, u) = (0.28716641518022373, 0.068777213178814, 0.009493712418100549)$$

and

$$D_{\text{face}} = 0.5233281002710417.$$

The minimum normalizer eigenvalue is 0.0577235542, so the point is well inside the positive region. Eight deterministic Nelder–Mead starts over all five fixed coordinates give values between

$$0.5233281002710413 \quad \text{and} \quad 0.5233281002710424.$$

The maximum apparent gain over the face is

$$6.661338147750939 \times 10^{-16}.$$

A finite-difference Hessian in the five-dimensional fixed space has sampled eigenvalues

$$(-2.4559, -1.4969, -1.0050, -0.6617, -0.3045),$$

all negative. This is strong, robust local evidence that the two residual directions do not improve the candidate. It is not a proof of global five-dimensional optimality.

P455 verdict: the five-dimensional fixed space and two-dimensional residual are **[Proven]**. Return to the O167 value is **[Strong evidence]**. The hypothesis that known symmetries alone force O167 is **[Refuted]**.

Chapter 6

Program P457 — six-decimal full-simplex erasure cover

6.1 Exact question

After P452 converted the P446 palindromic line into a theorem for the complete declared simplex, how much can the value bracket be tightened using the same auditable outward arithmetic without a new conceptual assumption?

6.2 Licensed one-dimensional reduction

P452 proves

$$D_{\text{coarse}}(p) \leq D_{\text{coarse}}\left(\frac{p + p^{\text{rev}}}{2}\right)$$

at $q = 4/5$, $\theta = 2\pi/15$. Every reversal average has the form

$$p(a) = (a, 1/2 - a, 1/2 - a, a), \quad 0 \leq a \leq 1/2.$$

P457 changes only the interval tolerance, from 10^{-3} to 10^{-6} . No full-simplex point is omitted because P452 supplies the exact reduction.

6.3 Finite cover

The directed binary64 implementation expands every arithmetic primitive by one representable number. Each branch cell receives an interval upper bound; midpoint evaluations update a certified feasible lower bound. The run uses

- 5,008 evaluated boxes;
- 5,009 terminal boxes;
- deterministic initial interval $[0, 1/2]$;
- no stochastic pruning.

The final bracket is

$$0.463278283192093 \leq D_{\text{coarse}}^{\text{global}} \leq 0.46327928294340853.$$

The width is

$$9.997513155113325 \times 10^{-7} < 10^{-6}.$$

The best certified point has

$$a = 0.2276921272277832,$$

and the conservative live maximizer hull is

$$[0.2178955078125, 0.234619140625].$$

The new gap is approximately 992.712 times smaller than the P452 gap.

P457 verdict: the full-simplex bracket is **[Computer-assisted proof]**. A closed-form maximizer, uniqueness, unrestricted inputs, adaptivity, and laboratory performance remain **[Open]**.

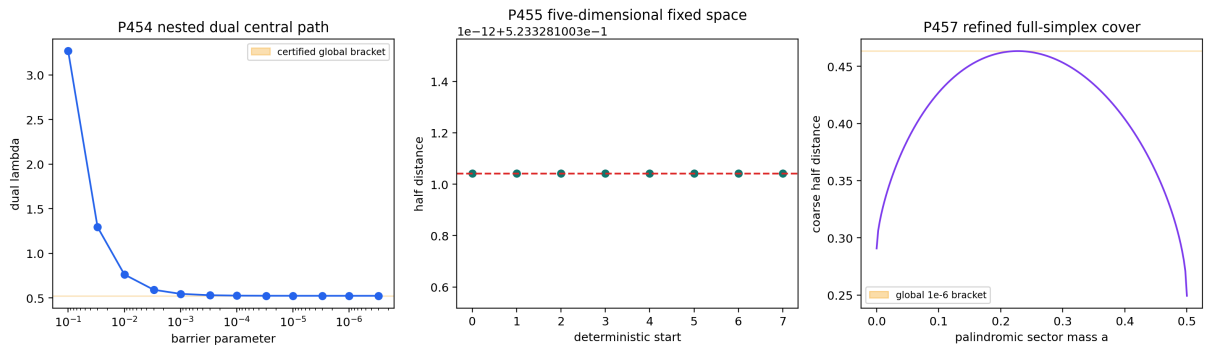


Figure 6.1: The nested-dual central path and certified global bracket, the five-dimensional ordered-symmetry audit, and the refined globally licensed coarse-erasure cover.

Chapter 7

Newly constructed theoretical objects

7.1 O170 — Nested Comb Dual Ladder

Required field	Definition and audit
Name	Nested Comb Dual Ladder
Domain/codomain	Three-slot primal tester SDP $\mapsto$ (X_3, X_2, X_1, λ) dual
Complete definition	The eight Loewner/scalar inequalities displayed in Section 4
Premises	P449 causal recursion; declared reduced Δ_3
Transformation law	Block levels follow the ordered causal trace hierarchy
Generated theorem	Global full-cone upper bound 0.523334700252
Necessity test	A scalar norm bound cannot resolve the P451 coherence gap to 10^{-5} scale
Removal attempt	Removing any level loses the corresponding causal normalization constraint
Strict/legacy relation	Downstream reduced channel; derives neither kernel and transfers no role
Selector dependence	Ordered slots are supplied; no QW-2191 discharge
Dimensional status	Dimensionless
Operational interpretation	Mathematical dual witness for binary causal discrimination
Failure mode	Does not prove exact optimizer or apparatus implementation
Confidence	[Proven] dual; [Computer-assisted proof] rational certificate

7.2 O171 — Ordered-Comb Symmetry Residual Space

Required field	Definition and audit
Name	Ordered-Comb Symmetry Residual Space

Required field	Definition and audit
Domain/codomain	Real $\mathcal{C}_3$ tangent space $\mapsto \ker(J_8(\cdot)J_8 - I)$
Complete definition	Exact rank-nine complement action on a fourteen-direction integer basis
Premises	Conjugation and global-complement invariance; causal order retained
Transformation law	Fixed under $N \mapsto J_8 N J_8$
Generated theorem	Fixed dimension five; O167 codimension two inside it
Necessity test	Explains why symmetry alone cannot close P451 exact globality
Removal attempt	Discarding the residual directions silently assumes two unproved constraints
Strict/legacy relation	Kernel-split robust downstream cone geometry
Selector dependence	Complement invariance distinguishes no orientation
Dimensional status	Dimensionless
Operational interpretation	Admissible ordered tester-normalization directions
Failure mode	Numerical non-improvement along residual directions is not a theorem
Confidence	[Proven] dimension; [Strong evidence] optimizer location

7.3 O172 — Six-Decimal Coarse-Erasure Cover

Required field	Definition and audit
Name	Six-Decimal Coarse-Erasure Cover
Domain/codomain	Palindromic $a \in [0, 1/2] \mapsto$ global objective bracket
Complete definition	Outward interval branch cover at tolerance 10^{-6}
Premises	P452 full-simplex reversal theorem and declared P446 channel/code
Transformation law	Reversal invariant
Generated theorem	Global bracket width $< 10^{-6}$
Necessity test	Reusing the old tolerance leaves a thousand-times wider uncertainty
Removal attempt	Floating maximization alone provides no global bound
Strict/legacy relation	Downstream, kernel-split robust after channel supply
Selector dependence	None; averaging does not select a direction
Dimensional status	Dimensionless; "six-decimal" is numerical, not a length scale
Operational interpretation	Mathematical code-performance certificate
Failure mode	Scope does not include arbitrary inputs, adaptive codes, or hardware

Required field	Definition and audit
Confidence	[Computer-assisted proof]

Chapter 8

Falsification and robustness audit

8.1 P454: no solver output was accepted as proof

The last central-path point fails the selected gradient tolerance. The report does not hide this. Its only role is to locate rational matrices. The theorem uses exact rational entries, exact leading principal minors, an upward-rounded objective, and a separately paid transcendental perturbation.

8.2 P454: global value is bounded, not identified exactly

The primal and dual witnesses differ by 6.6×10^{-6} . Nothing in duality says the O167 face attains the exact optimum until the gap is closed or complementary slackness is certified.

8.3 P455: symmetry shortcut is rigorously insufficient

The five-versus-three dimension count is exact. The two missing directions cannot be deleted because eight numerical starts did not exploit them. A new theorem must eliminate them or a future counterexample may use them.

8.4 P455: local Hessian is not global concavity proof

Although all five sampled Hessian eigenvalues are negative, the objective is concave in the normalizer but maximizing a concave function does not permit arbitrary conclusions from a sampled Hessian. A stationary supergradient or restricted dual is still required.

8.5 P457: expensive cover is finite and scoped

The 5,008-box run is deterministic and terminates. It does not justify repeating blind refinement indefinitely. The next conceptual move should isolate the derivative or prove uniqueness rather than merely add decimal places.

Chapter 9

Reproducibility boundaries

Program	Proof-grade layer	Scout-only layer	Remaining boundary
P454	Exact dual derivation, rational matrices, Sylvester margins, interval perturbation	Barrier path	Exact optimizer and uniqueness
P455	Exact dimension/rank/nullspace	Five-dimensional Nelder–Mead and Hessian	Eliminate or exploit two residual directions
P457	Outward exhaustive cover licensed by P452	1D Plot sampling	Closed form and uniqueness

The complete P457 batch takes approximately three minutes on the declared Intel i3-10110U system, dominated by interval evaluation. The P454 rational certificate is much cheaper than dense exact characteristic-polynomial isolation because Sylvester margins directly prove the required positivity.

Chapter 10

Physical and epistemic boundary

Question	Status after P454–P457
Mathematical three-slot tester value	Globally bracketed to $< 6.7 \times 10^{-6}$
Physical preparation of that tester	Not established
Physical clock or dimensional time	Not generated
Vertex POVM / twelve-state apparatus	Not demonstrated
Laboratory event record	None
QW-2191 selector	Open
Canonical orientation or phase origin	Open
Information-to-thermodynamic entropy	Not established
Legacy-to-strict completion	Open
Legacy physical-role transfer	Not started
Standard Model, gravity, L_{total} , ToE	Not established

The ordered causal hierarchy is a mathematical interface. It is not a physical arrow of time and does not source a Z12 orientation. The numerical parameter μ in the barrier method is not a physical scale.

Chapter 11

Ranked next research programs

Rank	Program	Exact target	Cheapest decisive method	Probability of decisive local result
1	P464	Close or sharply reduce the P454 6.6×10^{-6} primal–dual gap	Build a five-dimensional restricted dual/KKT inclusion using O171	
2	P458	Prove or refute uniqueness of the P457 palindromic maximizer	Interval isolate the derivative and certify monotonicity cells	
3	P459	Propagate P453 global uniqueness through the full O163 detector box	Combine forced support with P447 interval allocation sensitivities	
4	P460	Construct a null-cycle-quotient estimator or prove a no-go	Linear invariant analysis on the degree-11 moment quotient	
5	P461	Formalize the P452 coefficient theorem	Lean proof with a typed cosine-positivity provider boundary	
6	P462	Formalize P453 support forcing and Vandermonde uniqueness	Lean finite-measure complementarity plus finite Vandermonde lemma	
7	P463	Freeze an executable preparation contract for O169	Local schema, validator, synthetic integration test marked nonphysical	
8	P465	Produce a higher-precision rational O167 primal witness	Rationalize the face optimizer and repeat exact trace-distance certification	
9	P466	Test causal coherence at four slots	P449 recursion, global diagonal symmetry, one rational coherent witness	
10	P467	Generalize O170 to arbitrary slot number	Inductive primal/dual comb ladder theorem and dimension recurrence	

The selected next batch is

$$P464 \longrightarrow P458 \longrightarrow P459.$$

P464 attacks the most important remaining mathematical gap. P458 replaces further brute-force decimals with a uniqueness theorem or counterexample. P459 converts P453’s canonical mathematical gauge into a fully propagated, still explicitly nonphysical operational design.

Chapter 12

Exact artifact list

- FIN_Local_Research_Checkpoint_P454_P457_EN.md
 - FIN_Local_Research_Checkpoint_P454_P457_EN.tex
 - FIN_Local_Research_Checkpoint_P454_P457_EN.pdf
 - FIN_Current_Local_Research_Report_EN.pdf
 - FIN_Local_Research_Checkpoint_P454_P457_State.json
 - RELEASE_10_41_PROGRAMS_454_455_457.md
 - FIN_Checkpoint_P454_P457_AGENTS_Guardrail.txt
 - FIN_PROGRAMS_454_455_457_RELEASE_MANIFEST.sha256
 - FIN_Programs_454_455_457_Results.json
 - FIN_Programs_454_455_457_Summary.csv
 - FIN_Programs_454_455_457_P454_Nested_Dual.csv
 - FIN_Programs_454_455_457_P454_Rational_Dual.npz
 - FIN_Programs_454_455_457_P455_Symmetry_Residual.csv
 - FIN_Programs_454_455_457_P457_Refined_Cover.csv
 - FIN_Programs_454_455_457_Figures/p454_p457_dual_symmetry_and_refined_cover.png
 - fin_programs_454_455_457.py
 - fin_programs_454_455_457_to_latex.py
 - test_fin_programs_454_455_457.py
 - test_fin_checkpoint_p454_p457.py
-

Chapter 13

Restart instructions

Run the bounded scientific batch:

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 python3 fin_programs_454_455_457.py
```

Run current and inherited regression tests:

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 python3 -m unittest -v \  
  test_fin_programs_454_455_457.py \  
  test_fin_checkpoint_p454_p457.py \  
  test_fin_checkpoint_p451_p453.py
```

Regenerate and compile the monograph:

```
python3 fin_programs_454_455_457_to_latex.py  
lualatex -interaction=nonstopmode -halt-on-error \  
  FIN_Local_Research_Checkpoint_P454_P457_EN.tex  
lualatex -interaction=nonstopmode -halt-on-error \  
  FIN_Local_Research_Checkpoint_P454_P457_EN.tex
```

Verify the manifest:

```
sha256sum -c FIN_PROGRAMS_454_455_457_RELEASE_MANIFEST.sha256
```

Before P464, reread the latest AGENTS.md guardrail and do not promote the P455 numerical return to O167 into a symmetry theorem.

Chapter 14

Final conclusion

P454 changes the status of the three-slot causal problem from "one good coherent witness" to a globally bracketed finite SDP. P455 then explains why the remaining tiny gap cannot be erased by invoking the currently known symmetries: two exact ordered-comb directions survive outside O167. P457 shows that the separate coarse-erasure code problem is now controlled to six decimal places on its entire declared simplex.

The strongest interpretation surviving falsification is still purely mathematical: FIN supplies finite dimensionless operator, comb, and moment optimization problems that admit increasingly sharp primal, dual, symmetry, and variational certificates. None of these certificates supplies the missing bridge from mathematical information dynamics to experimentally testable dimensional physics.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 \  
python3 fin_programs_454_455_457.py  
  
MPLCONFIGDIR=/tmp/fin-mpl-1038 \  
python3 -m unittest -v \  
    test_fin_programs_454_455_457.py \  
    test_fin_checkpoint_p454_p457.py  
  
python3 fin_programs_454_455_457_to_latex.py  
lualatex -interaction=nonstopmode -halt-on-error \  
    FIN_Local_Research_Checkpoint_P454_P457_EN.tex  
  
sha256sum -c FIN_PROGRAMS_454_455_457_RELEASE_MANIFEST.sha256
```

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